

PRINCESS CROAK'S PALACE · PRINCESS CROAK · EVERY UNIT, EVERY DIFFICULTY

What every gnome, fairy, wasp, frog and Princess Croak herself does to your men on EASY, MEDIUM, HARD and MEGA, with every body's HP and head-count · read out of the live game v29.25 on Nov 2, 2026 14:04 EST · every damage number checked on the game's own machines (Chrome 252/252 · WebKit 252/252) · for Raziya

The one thing to know: in the palace **almost every hit takes BAR HITS off a man, not a KO**: her harp shot (**2 bar hits**, MEGA **3**), the guard's jab (1), the knight's jab (2), the slinger's glob (1), the wasp's sting (1, and the man is held 2 s), the hornet's sting (3, and he is bowled over), the frog's splat and tongue (1 each), and the sage's fire (1 a beat, 10 beats a ring). **Since v29.17 every one of those blows lands at 50 % on your men (THE DAMPER).** A level-70 rifleman has 7 bar hits (33 HP), so her harp KOs him on the **8th shot** on EASY to HARD and the **6th on MEGA**. Three things are a KO whatever a man's HP: **the gnome guard's RAM** (a charge at a man ahead within 300 px; a jeep or the tank is destroyed on the spear; the saber turns it about half the time, THE RIOT SHIELD's 5 and the Claymore's counted block eat it), **her LIGHT ARROWS** (MEDIUM up, at half her bar; MEGA again at a quarter: every man a bolt crosses, through every save: the one hit nothing turns), **her FACE SHOT** (a Sarge within 84 px: through the saber's block; the riot shield and the Claymore's block eat it, the deflect dome turns it) — and the sage's toad curse takes Sarge out of the fight for 5 s. Difficulty changes her bar (×2 · ×2.4 · ×6 · ×9), her lives (1 · 1 · 3 · 3), her cadence, the court's size and HP, and brings the arrows, the knights, the sages and the hornets.

THE BOSS LAB fights the palace at **LEVEL 70 · 77 · 83 · 90** (EASY · MEDIUM · HARD · MEGA): Sarge with the saber, the scout jeep, the dogfighter, the tank, the bomber and seven men at that rank (rifleman, M. gunner, bazooka, mortar, medic, flame, sweeper). The radio man, grenadier, commando and spy are a tap away in THE PLATOON. Sarge wears the two rank-5 gifts (telekinesis on the saber; the riot shield answers only with the shotgun in hand). Every "level" number on this page is that army.

What changed recently (the palace's own cards)

live since	when	card	what it does in this fight
v29.17	Nov 1	**CARD-DMP · THE DAMPER**	Every enemy blow on your men lands at 50 % in the campaign and the boss fights, the Boss Lab included (the battleground 70 %). A palace bar hit now lands 50 % × (90 % + 10 % × bar ÷ HP) of its weight, so a man stands about twice the blows: a level-70 rifleman falls to her 8th harp shot on EASY (the 4th before) and to a guard's 16th jab (the 8th). The ram, the light arrows and the face shot are KOs and land as shipped.
v28.93	Oct 31	**CARD-KCF · THE ELUSIVE CROAK**	The King Croak hero carries more HP: at level 70 he has 217 HP and a bar of 177 (the heroes' table).
v28.74	Oct 31	**CARD-HP3 · THE HEAVIES**	+3 HP every tenth rank for the bazooka, grenadier, flame and commando. At level 70: bazooka 66, grenadier 61, flame 80, commando 75 HP. Every palace hit counts in bar hits, so more HP means each bar hit is worth more HP, not fewer hits.
v28.68	Oct 30	**CARD-BRB · THE BOMBER**	Your bomber flies in every boss fight now, the palace included.
v28.65	Oct 30	**CARD-LRB · ONE HP NUMBER**	Your man's HP is the number on his unit sheet. A palace hit still counts in bar hits, but lands a little lighter on tough men: 90 % + 10 % × bar ÷ HP of its weight (a level-70 rifleman takes 0.92 of 2 from her harp, a flame 0.91).
v28.46	Oct 30	**CARD-BLP · THE BOSS LAB'S STOCK ARMY**	The lab fights the palace with its own army at LEVEL 70 · 77 · 83 · 90 (EASY · MEDIUM · HARD · MEGA): the one this page reads.
card 1213	—	**CARD-LA2 · THE KNIGHT'S MIRROR**	A knight's shield met from the front by a Sarge's own light arrow (the sniper mod won on HARD) throws it back at the shooter.
card 1027	—	**CARD-ML-A3 · PRINCESS CROAK, THE SIXTH HERO**	Beat her on MEGA and she joins as a hero (her HP row sits among the heroes above).

live since	when	card	what it does in this fight
card 1023	Oct 23	**CARD-PCS · PALACE SPACING**	Laura's spacing pass on her palace picture (the look; no fight numbers).
10/10	Oct 10	**CARD-PC2 · THE BIGGER COURT**	Wyatt: "give her more troops coming in" and "There should never be two Princess Croaks out at once": HARD and MEGA field more guards, slingers, wasps, knights, frogs; two hornets on HARD, three on MEGA; three sages at MEGA's open; THE COURT RETURNS ×8 on HARD and MEGA; one princess at every seat count.
card 1014	—	**THE PRESSURE · THE PALACE TUNE**	Her harp shot per tier (the cadence 2.4 · 2.0 · 1.8 · 2.0 s, MEGA 3 bar hits), the gate clocks, the knights at 75 %, the guard turning out at 50 % and 30 %, THE COURT RETURNS on the clock, the court's HP cells.
CARD-P3	—	**THE RETREAT · THE BUBBLE COURT**	HARD and MEGA: three bars; at a zero with bars left she flies back to her stage and calls her fairies and guards; the bar clock; the big guards charge at ×2.2.
card 1032	Sep 14–15	**CARD-PM-A · THE THRONE ROOM**	Wyatt: "Princess croaks map also needs to be redesigned to feel more epic". Six marble columns as cover, two frog fountains, the water stairs as a river with two bridges, the pool, her throne and harp behind her top step; MEGA moonlit.
card 0947e	Sep 11	**THE PALACE RAID**	Wyatt: "at half HP point, is going to begin to charge her light arrow… three large light arrows… if a commando has his reflect, you may reflect it right back at her… built-in auto-kill mechanics for any Sarge that comes too close". The fight as designed.
—	—	**THE FIRST READ**	The first read (Oct 31, v28.88) was built on Maker 1 from a shallow clone of the Big Mac's main 04d094bb69, with no git log there: the rows from v28.74 down were read off the client's own comments and the other kits' records, with a version where one is known. This re-read (THE DAMPER) is the Big Mac's, on its own client.

How a hit is counted

Your men carry **HP**: the number on their unit sheet (a level-70 flame has 80, a rifleman 33). Their health bar is also cut into a few **bar hits** (7 for most men at level 70–90, 6 for the medic and spy, 8 for the commando, 12 for the saber Sarge, 23 for the Claymore). Three kinds of hit in this fight:

- **Bar hits** (everything the court does, and her harp): a hit of N bar hits lands 90 % + 10 % × bar ÷ HP of N since v28.65, and since v29.17 50 % of that (THE DAMPER, below), so a level-70 rifleman takes 0.92 of his 7 from her harp (4.3 HP of 33) and 0.46 from a guard's jab. Bar hits mend after 4 s without a hit. A hero's bar IS his HP pool (a bar hit lands on it with no share, at the damper's 50 %).
- **THE DAMPER** (since v29.17): every blow an enemy lands on your men in the campaign and the boss fights, the Boss Lab included, lands at **50 %** of its size; it rides every bar hit on this page, the heroes' too, and every count and level here is counted with it. A KO hit is not dampened: it lands as shipped.
- **KO hits** (the guard's ram, her light arrows, her face shot): out in one hit whatever his HP. Only the arrows go through every save. Her face shot goes through the saber's block (the shot is sent with the saber's roll switched off: the KO door's rolled flag, by design, the auto-kill for a Sarge who comes too close), but THE RIOT SHIELD (5 a battle, the shotgun) and the Claymore's counted block (1 a battle: THE DUELIST, every Claymore's default spec; three only with FEAR or WALL picked) eat it whole, and the deflect dome turns it: measured, the shotgun Sarge fell to the 6th face shot and the Claymore to the 2nd. The ram reaches that door with no roll spent, so every save answers it: the saber turns it about half the time (measured 10/8 · 7/3 · 7/7 · 7/9 of 16 fresh Sarges KO'd on E · M · H · G, Chrome/WebKit), the plate and the counted block eat it (the shotgun Sarge fell to the 6th ram, the Claymore to the 2nd), the pet shield, the blue wall and the teal turn it; the deflect dome does not (its KO door catches only the officer's potshot and her face shot).
- **No hit**: the fairy mends her court and bubbles a guard BIG; the bubble itself pops on a green round; the sage's curse turns Sarge into a toad for 5 s (no fire, no swing).

Saves still roll on every bar hit: a jetting, dropping or dragged man cannot be touched, the pet shield, the deflect dome, **the saber Sarge's block turns a bar hit 50 % of the time** (measured: 36 of 75, 44 of 91, 26 of 50, 16 of 33 of her shots on the four tiers), **the shotgun Sarge's riot shield eats the first 5 hits whole** (measured), and the Claymore's counted block eats 1 (measured; THE DUELIST's one block is every Claymore's default spec, the table's 3 only with FEAR or WALL picked). The same saves answer the guard's ram, all but the deflect dome (its KO door catches only the officer's potshot and her face shot); her face shot skips only the saber's roll; her light arrows skip them all. A charged commando catches a light arrow and sends it back at her.

The strikes · what a hit does by difficulty

hit	EASY	MEDIUM	HARD	MEGA	who it hits	since
HER HARP SHOT · bar hits	2	2	2	3	the nearest man within 400 px, every 2.4 · 2 · 1.8 · 2 s (a level-70 rifleman: 4.3 HP a shot on EASY, 8.1 on MEGA)	the per-tier cadence and MEGA's third bar hit: card 1014
HER FACE SHOT	KO	KO	KO	KO	a Sarge within 84 px: the hop back and a shot through his saber block (THE RIOT SHIELD's 5 and the Claymore's counted block eat it; the deflect dome turns it)	unchanged
HER LIGHT ARROWS	—	KO	KO	KO ×2	every man a bolt crosses (three lanes 120 px apart, 56 px either side); a charged commando, the blue wall or the teal catches one	MEGA's second volley: card 1014
THE GNOME'S RAM	KO	KO	KO	KO	a man ahead within 300 px; a jeep or the tank is destroyed on the spear; the saber turns it about half the time, THE RIOT SHIELD's 5 and the Claymore's counted block eat it, the pet shield, the blue wall and the teal turn it (the deflect dome does not)	unchanged
THE GNOME'S JAB · bar hits	1	1	1	1	a man within 28 px	unchanged
THE KNIGHT'S JAB · bar hits	—	2	2	2	a man within 28 px	unchanged
THE SLINGER'S GLOB · bar hits	1	1	1	1	a man within 280 px	unchanged
THE SAGE'S FIRE · bar hits a beat	—	1	1	1	your three rearmost men, 10 beats over 5 s	unchanged
THE SAGE'S TOAD CURSE	—	5 s a toad	5 s a toad	5 s a toad	a Sarge within 200 px: no fire, no swing, hops at ×0.6	unchanged
THE WASP'S STING · bar hits	1	1	1	1	every man a dive crosses; held 2 s	unchanged
THE HORNET'S STING · bar hits	—	—	3	3	every man a dive crosses; bowled over	two hornets on HARD, three on MEGA: CARD-PC2
THE FROG'S SPLAT · TONGUE · bar hits	1 · 1	1 · 1	1 · 1	1 · 1	a man within 150 px (the hop) · 56 px (the tongue)	unchanged
THE FAIRY'S BUBBLE	BIG gnome: ×4 HP	BIG gnome: ×4 HP	BIG gnome: ×4 HP	BIG gnome: ×4 HP	her own court, never your men	unchanged

The blows as the game ships them. On your men every one lands at 50 % (THE DAMPER, since v29.17: the campaign and the boss fights, the Boss Lab included), times the share; the KOs land as shipped.

Who survives · the lab's level

How many of that hit in a row (inside the 4 s mend) KO a fresh man at the lab's level for that tier (EASY 70 · MEDIUM 77 · HARD 83 · MEGA 90), each blow landing at the damper's 50 %. **KO** = one hit. Hover a cell for the numbers.

KO one hit · 2 two · 3 three · 4 four · 5+ five or more

· a dot in the corner = the count measured on the game's own machines in this read (both engines agree) · +½ = the saber turns about half (her harp shot, the ram) · +5 = the riot shield eats 5 first · +1 on the Claymore = his counted block eats 1 first · "·" = not in this fight / only a Sarge

man	HP E · M · H · G	bar hits	HER HARP SHOT				GNOME'S JAB				KNIGHT'S JAB				SLINGER'S GLOB				WASP'S STING				HORNET'S STING				FROG'S SPLAT · TONGUE				SAGE'S FIRE (beats)				GNOME RAM	LIGHT ARROWS				FACE SHOT
			E	M	H	G	E	M	H	G	E	M	H	G	E	M	H	G	E	M	H	G	E	M	H	G	E	M	H	G	E	M	H	G		E	M	H	G	
THE LAB'S STOCK ARMY (LEVEL 70 · 77 · 83 · 90)																																							
RIFLEMAN	**33 · 33 · 37 · 41**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
M. GUNNER	**50 · 50 · 55 · 60**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
BAZOOKA	**66 · 66 · 75 · 83**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
MORTAR	**30 · 30 · 33 · 37**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
MEDIC	**29 · 29 · 33 · 36**	6	7	7	7	5	14	14	14	14	·	7	7	7	14	14	14	14	14	14	14	14	·	·	5	5	14	14	14	14	·	14	14	14	KO	·	KO	KO	KO	·
FLAME	**80 · 80 · 90 · 100**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
SWEEPER	**31 · 31 · 35 · 38**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
SARGE · SABER	**40 · 41 · 44 · 47**	12	13+½	13+½	13+½	9+½	26	26	26	26	·	13	13	13	26	26	26	26	26	26	26	26	·	·	9	9	26	26	26	26	·	26	26	26	KO+½	·	KO	KO	KO	KO
A TAP AWAY (THE PLATOON, SARGE'S OTHER IRONS)																																							
RADIO MAN	**31 · 31 · 35 · 38**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
GRENADIER	**61 · 61 · 68 · 76**	7	8	8	8	6	16	16	16	16	·	8	8	8	16	16	16	16	16	16	16	16	·	·	6	6	16	16	16	16	·	16	16	16	KO	·	KO	KO	KO	·
COMMANDO	**75 · 75 · 84 · 93**	8	9	9	9	6	18	18	18	18	·	9	9	9	18	18	18	18	18	18	18	18	·	·	6	6	18	18	18	18	·	18	18	18	KO	·	KO	KO	KO	·
SPY	**36 · 36 · 41 · 45**	6	7	7	7	5	14	14	14	14	·	7	7	7	14	14	14	14	14	14	14	14	·	·	5	5	14	14	14	14	·	14	14	14	KO	·	KO	KO	KO	·
SARGE · PISTOL	**41 · 41 · 44 · 47**	11	12	12	12	8	24	24	24	24	·	12	12	12	24	24	24	24	24	24	24	24	·	·	8	8	24	24	24	24	·	24	24	24	KO	·	KO	KO	KO	KO
SARGE · SHOTGUN	**40 · 41 · 44 · 47**	8	9+5	9+5	9+5	6+5	18	18	18	18	·	9	9	9	18	18	18	18	18	18	18	18	·	·	6	6	18	18	18	18	·	18	18	18	KO+5	·	KO	KO	KO	KO+5
SARGE · SNIPER	**31 · 31 · 34 · 37**	8	9	9	9	6	18	18	18	18	·	9	9	9	18	18	18	18	18	18	18	18	·	·	6	6	18	18	18	18	·	18	18	18	KO	·	KO	KO	KO	KO
SARGE · CLAYMORE	**71 · 73 · 73 · 76**	23	25+1	25+1	25+1	17+1	50	50	50	50	·	25	25	25	50	50	50	50	50	50	50	50	·	·	17	17	50	50	50	50	·	50	50	50	KO+1	·	KO	KO	KO	KO+1

The bar is 7 for most men from level 70 to 90, so the counts hold across the tiers; HP is what each bar hit is worth. HER HARP SHOT: every man and iron measured on every tier, each to his KO (the dots). The court's hits were measured on a rifleman (jab, glob, sting, frog, fire), a flame (knight, hornet, fire) and a grenadier (fire): each hit's weight on every tier, and the count to the KO (the dot) for the guard's jab, the slinger's glob and the knight's jab; the rest of the column is the same arithmetic. SAGE'S FIRE counts beats in a row; a ring holds 10 (5 s), so a count over 10 means he walks out of the first ring: measured on MEDIUM (level 77), the rifleman with 4.61 of his 7, the flame with 4.54 of his 7, the grenadier with 4.56 of his 7; the 4 s mend gave 1 back before the next ring (12 s on), which KO'd each on its 8th beat (18 beats in all). THE RADIO MAN's harp cells hold his own call (a paratrooper every 15 s alive), so they count her harp on him. With the call live, as in a fight, the paratrooper draws her next shots while his bar mends: measured EASY 14 landed in 52.4 s to his KO, against the 8 held (MEDIUM, HARD, MEGA: he falls first). GNOME RAM, LIGHT ARROWS and FACE SHOT: one hit, whatever the HP; only the arrows pass every save (the ram was measured on a rifleman, a flame and Sarge's saber, pistol, shotgun and Claymore; the face shot on all four irons). SARGE · SABER's harp cells carry no dot: 13 · 13 · 13 · 9 is the count if every landed shot came inside the 4 s mend, but his block spaces them and his bar mends between, so the machines measured 39/62 · 47/27 · 24/24 · 17/19 landed shots to his KO (E · M · H · G, Chrome/WebKit, after 180/309 · 183/102 · 91/92 · 66/88 s: at the damper's half his mend nearly keeps pace with her).

The heroes at the lab's level (one in command), measured

hero	HP (E · M · H · G)	bar hits	her harp shot	shots to KO	measured: shots landed in 60 s (Chrome/WebKit where they differ)	her face shot	her light arrow
KING YETI	**138 · 139 · 176 · 213**	126 · 126 · 182 · 238	1 · 1 · 1 · 1.5 of his bar a shot	126 · 126 · 182 · 159	24 · 29 · 32 · 29	never: not a Sarge	KO by the code (the same door as a man's); not staged here
KING REX	**209 · 211 · 231 · 239**	137 · 137 · 163 · 169	1 · 1 · 1 · 1.5 of his bar a shot	137 · 137 · 163 · 113	25 · 29 · 33 · 29	never: not a Sarge	KO by the code (the same door as a man's); not staged here
KING CROAK	**217 · 219 · 226 · 233**	177 · 177 · 183 · 189	1 · 1 · 1 · 1.5 of his bar a shot	177 · 177 · 183 · 126	25 · 29 · 33 · 29	never: not a Sarge	KO by the code (the same door as a man's); not staged here
JANE	**126 · 128 · 135 · 142**	68 · 68 · 74 · 80	1 · 1 · 1 · 1.5 of his bar a shot	68 · 68 · 74 · 54	25 · 29 · 33 · 29	never: not a Sarge	KO by the code (the same door as a man's); not staged here
EMPEROR BEAR	**218 · 221 · 241 · 249**	137 · 137 · 163 · 169	1 · 1 · 1 · 1.5 of his bar a shot	137 · 137 · 163 · 113	20/22 · 23/16 · 29/27 · 18/21	never: not a Sarge	KO by the code (the same door as a man's); not staged here
PRINCESS CROAK	**97 · 98 · 104 · 111**	77 · 77 · 83 · 89	1 · 1 · 1 · 1.5 of his bar a shot	77 · 77 · 83 · 60	21 · 21 · 25/23 · 23	never: not a Sarge (measured at 60 px)	KO by the code (the same door as a man's); not staged here

A hero's bar is his HP pool: her harp takes 1 (MEGA 1.5) of it a shot: her 2 (MEGA 3) at the damper's 50 %, with no share (measured on every tier). No hero fell inside 60 s on any tier (the damper halves her shot on the pool too). A hero walks, so the count landed in 60 s is the engines' clocks as much as hers: EMPEROR BEAR on EASY (20 on Chrome, 22 on WebKit); EMPEROR BEAR on MEDIUM (23 on Chrome, 16 on WebKit); EMPEROR BEAR on HARD (29 on Chrome, 27 on WebKit); EMPEROR BEAR on MEGA (18 on Chrome, 21 on WebKit); PRINCESS CROAK on HARD (25 on Chrome, 23 on WebKit); every shot's weight was the same on both. Her face shot is for a Sarge alone (measured on the Princess Croak hero). The guard's ram and her light arrows go through the same doors for a hero as for a man (the code; not staged on this page).

The level a man first survives

For her harp shot and the hornet's sting (the hits that take part of a man). The KO hits have no such level: no rank survives them. **Green** = he already does at the lab's EASY level 70; — = not even at rank 100.

			HER SHOTS IN A ROW · EASY–HARD (2 bar hits)				… MEGA (3)				HORNET STINGS		HER SHOTS TO KO · EASY–HARD, BY RANK						
man	HP / bar at 0	HP / bar at 100	1	2	3	4	1	2	3	4	1	2	r0	r10	r25	r50	r70	r90	r100
RIFLEMAN	3 / 3	45 / 8	0	0	2	5	0	2	10	19	0	2	3	6	7	8	8	8	9
M. GUNNER	8 / 3	65 / 8	0	0	0	5	0	0	10	19	0	0	4	6	7	8	8	8	9
BAZOOKA	3 / 3	92 / 8	0	0	2	6	0	2	10	19	0	2	3	6	7	8	8	8	9
MORTAR	3 / 3	40 / 8	0	0	2	6	0	2	10	19	0	2	3	6	7	8	8	8	9
MEDIC	2 / 2	40 / 7	0	3	8	10	0	8	19	42	0	8	2	5	6	7	7	7	8
FLAME	6 / 3	110 / 8	0	0	0	5	0	0	10	19	0	0	4	6	7	8	8	8	9
SWEEPER	3 / 3	42 / 8	0	0	2	6	0	2	10	19	0	2	3	6	7	8	8	8	9
RADIO MAN	3 / 3	42 / 8	0	0	2	6	0	2	10	19	0	2	3	6	7	8	8	8	9
GRENADIER	3 / 3	84 / 8	0	0	2	6	0	2	10	19	0	2	3	6	7	8	8	8	9
COMMANDO	7 / 4	102 / 9	0	0	0	0	0	0	5	10	0	0	5	7	8	9	9	9	10
SPY	2 / 2	50 / 7	0	3	8	10	0	8	19	42	0	8	2	5	6	7	7	7	8
SARGE · SABER	9 / 8	50 / 13	0	0	0	0	0	0	0	0	0	0	9	11	12	13	13	13	15

			HER SHOTS IN A ROW · EASY–HARD (2 bar hits)				… MEGA (3)				HORNET STINGS		HER SHOTS TO KO · EASY–HARD, BY RANK						
man	HP / bar at 0	HP / bar at 100	1	2	3	4	1	2	3	4	1	2	r0	r10	r25	r50	r70	r90	r100
SARGE · PISTOL	8 / 4	50 / 12	0	0	0	0	0	0	5	10	0	0	5	7	8	12	12	12	13
SARGE · SHOTGUN	9 / 4	50 / 9	0	0	0	0	0	0	5	10	0	0	5	7	8	9	9	9	10
SARGE · SNIPER	6 / 4	40 / 9	0	0	0	0	0	0	5	10	0	0	5	7	8	9	9	9	10
SARGE · CLAYMORE	22 / 19	78 / 24	0	0	0	0	0	0	0	0	0	0	20	23	24	25	25	25	26

Read from the game's own HP and bar ladders at every rank 0–100 (the tenth-rank extras included; no gifts, soul or gear). Sarge's rows read the iron at the same rank. HER SHOTS TO KO by rank on EASY to HARD (2 bar hits a shot, landing at the damper's 50 %); MEGA's third bar hit KOs in fewer (a level-90 rifleman: 6th shot, the 8th at level 70 on EASY).

The units

unit	attack	what a hit does	HP EASY	MED	HARD	MEGA	n EASY	MED	HARD	MEGA	leak	tags
 **PRINCESS CROAK (the boss)** walks 24 px/s, keeps 240 px off your line and backs off at 96 px/s inside 160 px · lives 1 · 1 · 3 · 3	**HARP SHOT · FACE SHOT · LIGHT ARROWS** a harp shot at the nearest man within 400 px every 2.4 · 2 · 1.8 · 2 s · a Sarge within 84 px: the face shot · at half her bar (MEDIUM up) the three light arrows	**2 · 2 · 2 · 3 bar hits** a harp shot (E · M · H · G) · the face shot and the light arrows are a **KO**	320	384	960	1440	1	1	1	1	8	400
 **THE GNOME GUARD** walks 26 px/s · his shield turns every 2nd round from the front; bullets do half, blasts double · a fairy's bubble makes him BIG (×4 HP) · Princess Croak's summoned guards charge the line at ×2.2 once big	**RAM · JAB** braces 0.45 s then charges 380 px/s at a man ahead within 300 px, 100 px past him, then rests 1.6 s · jabs a man within 28 px every 1 s (0.2 s wind-up)	the ram a **KO** unless a save answers (the saber about half the time, THE RIOT SHIELD's 5, the Claymore's counted block; a jeep or the tank destroyed on the spear) · the jab **1 bar hit**	8	10	12	14	20	27	72 +3 a call	94 +4 a call	1	18
 **THE GNOME KNIGHT** MEDIUM up · walks 20 px/s · his shield turns EVERY round from the front (shoot him from behind or the flank); bullets do half, blasts double · a shield met by a light arrow from a Sarge's mod throws it back	**JAB** jabs a man within 28 px every 1.4 s (0.35 s wind-up); no ram	**2 bar hits**	—	18	21	25	—	2	4	5	2	36

unit	attack	what a hit does	HP EASY	HP MED	HP HARD	HP MEGA	n EASY	n MED	n HARD	n MEGA	leak	tags
 **THE GNOME SLINGER** walks 24 px/s	**GLOB** aims 0.6 s then slings at a man within 280 px, every 3 s and the aim (measured 3.1–3.6 · 3–3.5 · 3–3.6 · 3–3.6 s) · backs off from a man inside 120 px	**1 bar hit**	4	5	6	7	2	3	6	7	1	12
 **THE GNOME SAGE** MEDIUM up (one at the pond; HARD three; MEGA three at the open and two at the pond) · walks 22 px/s	**FIRE RINGS · TOAD CURSE** the first fire at 6 s then every 12 s: the staff up 0.6 s, the rings 1.0 s, then 3 patches 56 px around their centres (112 px across) · the curse on a Sarge within 200 px, a 0.6 s cast, every 15 s	the fire **1 bar hit a beat**, a beat every 0.5 s for 5 s (10 beats) under each of your three rearmost men · the curse turns **Sarge** into a toad for 5 s (he hops at ×0.6, cannot fire or swing)	—	13	15	18	—	1	3	5	2	30
 **THE FAIRY** hovers 80 px behind the front man · Princess Croak's summoned fairies blow bubbles and never mend	**MEND · BUBBLE** the first bubble at 4 s, then every 10 s (0.6 s blow, 120 px/s); one big man a fairy at a time	no hit on your men: mends her court **1 HP a second** within 120 px and blows a bubble that makes the front gnome **BIG: ×4 HP, ×1.5 size, jabs ×1.5 as often**	3	4	5	5	1	2	2 +3 a call	2 +4 a call	1	14
THE BUBBLE a bubble gains on a man marching west; it stalls 12 px short of a man who stands still (measured)	**— (a buff in flight)** flies 120 px/s at the front gnome; BIG on arrival	none on your men; it pops on a green round (1 HP)	1	1	1	1	one a fairy every 10 s	one a fairy every 10 s	one a fairy every 10 s	one a fairy every 10 s	0	0
 **THE WASP** flies 78 px/s	**STING** dives 520 px/s at a man within 260 px, stings every man his pass crosses, climbs 2.2 s between dives (half that while a hornet is on the floor)	**1 bar hit** + the man is **HELD 2 s** (no march, no gun)	2	3	3	4	24	21	41	50	1	8

unit	attack	what a hit does	HP EASY	MED	HARD	MEGA	n EASY	MED	HARD	MEGA	leak	tags
 **THE HORNET** HARD and MEGA · flies 60 px/s · while he lives every wasp dives twice as often	**HEAVY STING** hangs 0.4 s over his man (the tell) then dives 420 px/s at a man within 300 px, climbs 2.6 s	**3 bar hits** + the man is **bowled over**	—	—	18	22	—	—	2	3	2	34
 **THE POND FROG** born from the pool's rim; never walks	**SPLAT · TONGUE** hops 460 px/s at a man within 150 px (one a hop, every 2 s) · the tongue at a man within 56 px every 2 s · beside a man: a tongue and a hop about every 3 s each (measured: 6 hits in 7 s, 2.76 bar hits landed)	the splat **1 bar hit** + a shove · the tongue **1 bar hit**	4	5	6	7	2	3	6	6	1	10

HP = each body as the palace's own wave rows bear it on that tier (measured in this read: the court ×1 · ×1.25 · ×1.5 · ×1.8, her bar ×2 · ×2.4 · ×6 · ×9); a BIG guard has four times his. n = bodies named in the solo wave sheet (one commander), THE COURT RETURNS rows included; her summons come on top ("+N a call"). leak = lives it costs at the toy box. tags = dog tags paid when it falls. Our rounds on the gnomes: a guard's shield turns every 2nd round from the front and bullets do half (blasts double); a knight's shield turns every round from the front (measured: 5 of 5 rifle rounds, his bar unmoved). What a hit does is the blow as the game ships it: on your men it lands at 50 % (THE DAMPER, since v29.17), times the share; a KO is not dampened.

Princess Croak, move by move

move	EASY	MEDIUM	HARD	MEGA	how it works
HARP SHOT	**2 bar hits** · every 2.4 s	**2 bar hits** · every 2 s	**2 bar hits** · every 1.8 s	**3 bar hits** · every 2 s	The nearest man within 400 px (a jetting, cloaked or dragged man is not picked). Measured every 2.42 · 2.02 · 1.82 · 2.02 s, the first 1.6 s after she steps on; a level-70 rifleman takes 0.92 of his 7 bar hits a shot (the damper lands 50 % of it) and falls to the 8th. She glides to 240 px from your line and backs off east at 96 px/s when a man closes inside 160 px.
FACE SHOT	**KO** · 0.47 s	**KO** · 0.47 s	**KO** · 0.47 s	**KO** · 0.47 s	A SARGE within 84 px (the Claymore, the pistol, any iron; never a man or a hero: measured). The 0.12 s crouch tell, the 80 px hop back (0.3 s), and the shot in the face: a KO sent with the saber's roll switched off (the KO door's rolled flag, by design: the auto-kill for a Sarge who comes too close), while THE RIOT SHIELD, Francis's block or the Claymore's counted block eat it whole and the deflect dome turns it. Measured: a saber Sarge at 60 px KO'd at 0.47 s on every tier; kept 60 px off her, the shotgun Sarge's plate ate 5 face shots and he fell to the 6th, the Claymore's block ate 1 and he fell to the 2nd; a rifleman at 60 px took the harp instead.
LIGHT ARROWS	off	**KO** · at 50 % of her bar	**KO** · at 50 % of her bar	**KO** · at 50 % of her bar and again at 25 %	MEDIUM and up: when her bar falls to half she roots and charges 2 s ("Let me show you true power."), then three bolts fly west from her, 120 px apart (her row and one above and below), 900 px/s: every man within 56 px of a bolt's line is KO'd through every save (a machine too); the bolts end at your deploy line, the toy box is never hit. Measured: three men in the three lanes KO'd 2 s after the gate, a medic 60 px off a lane untouched. MEGA fires it again at a quarter ("Punishment" in E Minor). On HARD and MEGA the arrow act also comes with her retreats (below).

move	EASY	MEDIUM	HARD	MEGA	how it works
THE CATCH	—	she takes **96 HP** (25 % of her bar)	she takes **240 HP** (25 % of her bar)	she takes **360 HP** (25 % of her bar)	A charged commando standing in a lane (or covering a man with THE BUDDY SHIELD), the blue wall, the teal's save or a swinging Claymore CATCHES a bolt: it flies back east at her, every gnome it crosses falls, and she takes a quarter of her bar (measured: the gnome in its path fell, her bar –25 %). A commando catches ONE bolt a fight (measured on MEGA: the second volley KO'd him).
THE GATES	75 %: GUARDS! TO ME! 50 %: THE GUARD TURNS OUT 30 %: THE POND! EVERYONE OUT OF THE POND! then THE COURT RETURNS ×5 every 40 s (2 gnome, 4 wasp)	75 %: GUARDS! TO ME! 50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS 30 %: THE POND! EVERYONE OUT OF THE POND! then THE COURT RETURNS ×5 every 36 s (3 gnome, 3 wasp)	75 %: GUARDS! TO ME! 50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS 30 %: THE POND! EVERYONE OUT OF THE POND! then THE COURT RETURNS ×8 every 24 s (6 gnome, 4 wasp)	75 %: GUARDS! TO ME! 50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS 30 %: THE POND! EVERYONE OUT OF THE POND! 25 %: AGAIN. — THE LIGHT ARROWS then THE COURT RETURNS ×8 every 20 s (8 gnome, 5 wasp)	Her court arrives as her bar falls (each gate opens on the clock too: 56 · 48 · 42 · 42 s after it comes up). The gates latch once; a later bar does not reopen them. THE COURT RETURNS rows come on the clock alone, every 40 · 36 · 24 · 20 s.
HER LIVES	**1** bar	**1** bar	**3** bars of 960	**3** bars of 1,440	EASY and MEDIUM: her first zero is her fall. HARD and MEGA: THE FALL THAT ISN'T — at zero with bars left she flies back to her stage (≤ 1 s), holds 0.8 s, rises with a whole new bar and acts: HARD "summon" then "arrow", MEGA "both" twice. Measured on HARD: the retreat at 0.9 s, the court at 2.2 s; the second zero brought the arrows and KO'd the rifleman in her middle lane. Her third zero is her fall.
THE BUBBLE COURT	—	—	**3 guards + 3 fairies**	**4 guards + 4 fairies**	"MY FAIRIES! Get big my darlings!": the guards appear at her stage at ×1.5 / ×1.8 HP, each fairy on her own guard; the bubbles make them BIG (×4 HP) and the big guards CHARGE your line at ×2.2 speed (measured: 3 of 3 big within 4 s on HARD). Her fairies never mend.
THE BAR CLOCK	—	—	every **60 s**	every **45 s**	A later bar that stands 60 s (HARD) / 45 s (MEGA) calls THE BUBBLE COURT again (measured: the call at 60 s and 45 s). The fight always moves.
THE WAVE AFTER HER FIRST BAR	—	—	+3 guards + 3 fairies	+4 guards + 3 fairies	Once her first bar is down every COURT RETURNS wave also fields charging guards with a fairy each (the code's own rows; not staged on this page).
WALK	24 px/s	24 px/s	24 px/s	24 px/s	Her glide; 96 px/s on the back-off. A press she cannot gain a pixel on for 1.2 s is given up for a wander step (the throne room's columns and fountains).
HER BAR (one commander)	**320**	**384**	**960**	**1,440**	160 HP × the tier's cell (×2 · ×2.4 · ×6 · ×9); HARD and MEGA three times over (lives). Her fall ends the fight. With more commanders the bar grows (the table below).
THE TOY BOX	–8 lives if she reaches it	–8 lives if she reaches it	–8 lives if she reaches it	–8 lives if she reaches it	Every other unit costs its LEAK (1–2) of the box's lives.

Her bar by the number of commanders

	1 commander	2 commanders	3 commanders	4 commanders	5 commanders	10 commanders
EASY	320	608	896	1,184	1,472	2,912
MEDIUM	384	730	1,075	1,421	1,766	3,494
HARD	960	1,824	2,688	3,552	4,416	8,736
MEGA	1,440	2,736	4,032	5,328	6,624	13,104

Read from the game's own wave sheet with 1, 2, 3, 4, 5 and 10 commanders in the fight. HARD and MEGA: three such bars (her lives). One Princess Croak at every seat count; her court scales with the party.

What changes from EASY to MEGA

	EASY	MEDIUM	HARD	MEGA
Her harp shot (bar hits · seconds between)	2 · 2.4	2 · 2	2 · 1.8	3 · 2
An enemy's blow on your men lands at (THE DAMPER)			50 %	
Her light arrows	—	at 50 %	at 50 %	**at 50 % and 25 %**
Her lives (bars)	1	1	3	3
Her bar (one commander)	320 (×2)	384 (×2.4)	960 (×6)	1,440 (×9)
The court's HP	×1	×1.25	×1.5	×1.8
Guards · slingers · knights · sages in the sheet	20 · 2 · — · —	27 · 3 · 2 · 1	72 · 6 · 4 · 3	94 · 7 · 5 · 5
Wasps · hornets · frogs · fairies in the sheet	24 · — · 2 · 1	21 · — · 3 · 2	41 · 2 · 6 · 2	50 · 3 · 6 · 2
Bodies in the sheet, solo (she is not counted)	49	59	136	172
Gate clock (a gated wave opens by itself after)	56 s	48 s	42 s	42 s
THE COURT RETURNS (waves × every)	5 × 40 s	5 × 36 s	8 × 24 s	8 × 20 s
THE BUBBLE COURT at a retreat (guards + fairies) · the bar clock	—	—	3 + 3 · 60 s	4 + 4 · 45 s
The room	day	day	day	**moonlit** (night)
The lab's stock army level	70	77	83	90

The wave sheets (solo, one commander)

EASY · 49 bodies · level 70

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   4 gnome guard, 2 slinger, 1 fairy · PRINCESS CROAK (bar 320)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 56 s))
   2 wasp, 1 pond frog

MEDIUM · 59 bodies · level 77

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   5 gnome guard, 3 slinger, 1 fairy · PRINCESS CROAK (bar 384)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 48 s))
   2 knight, 1 fairy, 4 wasp, 2 pond frog

HARD · 136 bodies · level 83

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   10 gnome guard, 6 slinger, 1 fairy · PRINCESS CROAK (bar 960)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 42 s))
   4 knight, 1 fairy, 7 wasp, 2 hornet, 3 pond frog

MEGA · 172 bodies · level 90

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   12 gnome guard, 7 slinger, 1 fairy, 3 sage · PRINCESS CROAK (bar 1,440)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 42 s))
   5 knight, 1 fairy, 8 wasp, 3 hornet, 3 pond frog

3. **THE GUARD TURNS OUT** (holds while princess stands above 50% (or 56 s))
   3 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 56 s))
   1 pond frog, 3 gnome guard
5. **GUARDS! TO ME!** (holds while princess stands above 0% (or 40 s))
   2 gnome guard, 4 wasp × 5

3. **LET ME SHOW YOU TRUE POWER.** (holds while princess stands above 50% (or 48 s))
   4 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 48 s))
   1 pond frog, 3 gnome guard, 1 sage
5. **GUARDS! TO ME!** (holds while princess stands above 0% (or 36 s))
   3 gnome guard, 3 wasp × 5

3. **LET ME SHOW YOU TRUE POWER.** (holds while princess stands above 50% (or 42 s))
   8 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 42 s))
   3 pond frog, 6 gnome guard, 3 sage
5. **GUARDS! TO ME!** (holds while princess stands above 0% (or 24 s))
   6 gnome guard, 4 wasp × 8

3. **LET ME SHOW YOU TRUE POWER.** (holds while princess stands above 50% (or 42 s))
   10 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 42 s))
   3 pond frog, 8 gnome guard, 2 sage
5. **AGAIN.** (holds while princess stands above 25% (or 42 s))
   nothing new: her second volley
6. **GUARDS! TO ME!** (holds while princess stands above 0% (or 20 s))
   8 gnome guard, 5 wasp × 8

She opens on her top step; her court walks on behind her over the first seconds. Each gated wave opens when her bar falls to its mark, or by itself after the gate clock. THE COURT RETURNS rows come on the clock alone, whatever her bar.

How this was read and checked

The live client v29.25 (index.html md5 93df2c286601f6a041969f40e7f12824, the Big Mac's own client at HEAD) was opened headless on this Mac and read with the game's own functions, THE DAMPER's table by name among them: every table, every tier's wave sheet, a real lab fight per tier for the stock army and its level, every man's HP and bar at every rank, her bar for 1 to 10 commanders. Then each hit was **measured on the real machines** in a BOSS LAB fight on every tier, every attacker born through the palace's own wave row so the tier's stamps ride: THE DAMPER in every lab fight (the game's own share on each tier equals its cell, and every hit on our men rode it); her harp shot on all eleven men and Sarge's five irons, each to his KO, and on the six heroes for 60 s; her face shot on a saber and a pistol Sarge (and that a rifleman, a hero and a Sarge at 100 px are spared); her light arrows on three men in the three lanes and a medic between them, the commando's catch and the ray thrown back at her and her gnome, MEGA's second volley; the guard's jab and ram (and the scout jeep, the V2 jeep and the tank on the spear), the knight's jab and shield, the guard's shield and armour under our rifle, the slinger's glob, the sage's toad curse and fire, the fairy's mend and bubble, the wasp's sting and the hold, the hornet's sting and the bowl-over, the frog's splat and tongue; on HARD and MEGA her retreat, THE BUBBLE COURT, the big guards' charge, the bar clock and the second zero; every body as the tier bears it. Chrome 252/252 · WebKit 252/252 checks passed, zero page errors. The two engines' READS are identical (checked), and every damage fact in the two proofs agrees (1388 facts checked across the engines: each hit's weight, every KO and every save, every body). What differs between Chrome and WebKit is dice and clocks, and this page prints both where it matters: the saber Sarge's harp shots landed to his KO (39/62 · 47/27 · 24/24 · 17/19 on E · M · H · G, after 180/309 · 183/102 · 91/92 · 66/88 s); the gnome's ram on 16 fresh saber Sarges (10/8 · 7/3 · 7/7 · 7/9 KO'd); EMPEROR BEAR's shots landed in 60 s (EASY 20/22, MEDIUM 23/16, HARD 29/27, MEGA 18/21); PRINCESS CROAK's shots landed in 60 s (HARD 25/23); the frame a hit lands on, and the slinger's wander.

Sources (the code's own names): FDEF (princess · gnome · gslinger · gsage · gknight · fairy · fbubble · wasp · hornet · pfrog); updateFoePrincess (PC_SHOT, the halt and back-off, pcNearest, pcFaceShot → knock(rolled), the gates → pcFire), the 'lightarrow' shot branch (knockKill, cmdoCharged/cmdoCovers, tkDeflT, mblRay, arrowBack → hurt under TK_REFL), PC_ARROW_TIER, PC_LIVES, pcRetreat / pcRetTick (PC_RET, PC_RET_ACTS), pcSummon / pcSummonIn (PC_SUMMON, PC_COURT_HPX), PC_BAR_CLK, pcwWave (PCW), PC_GIANT_SPDX; updateFoeTrike (the ram bag → knock; the vehicle branch → knockKill), gnJabTick, gnSlingTick + tanShoot (def.pips → pipN), gnCasterTick (fsIgnite → fsTick: tanHit fire beats; toadT), gnFairyTick / gnBubbleTick (big, bigX), updateFoeSwoop (sting.pips → tanHit, kyHeldT, ggShove), updateFoePounce (pounce.pips, tongue), hurt (the gnomes' shield shN, gnArmorX); tanHit, DMG_DAMP / dmpShare / dmpX (THE DAMPER inside tanHit's pip write), DSM (k 0.9), dsmH / dsmX, dhtH, gritCap, radioTick (RADIO_DROP_T, held in the harp staging), mgCtr, enhSoakM, saberBlocks, sargeAbsorbs (ENH_SHIELD_N), cmdAbsorbs; raidCfg's palace branch (PC_HPX, the court's cells, hpGate rows with clk, PC_GATE_CLK, PC_RETURN_CLK), blbSheet('palace', tier, N), blbStockDef (BLB_STOCK_LV.palace), blbLaunch; bfHpAt, usBfHP, sgBfHP, heroBfHP. Kit: design/boss-princess-unit-breakdown-1031/ (kit/readPalace.mjs · kit/provePalace.mjs · kit/build.py · kit/render.mjs · kit/export_dash.py; rig/lrbShared.mjs; data/; proof/logs/). Zero game bytes.